

Program.1- Implementation of decision tree.

Import necessary libraries

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
```

Step 1: Load the dataset

We will use the Iris dataset, a simple dataset for classification problems

```
iris = load_iris()
X = iris.data # Features (flower measurements)
y = iris.target # Labels (flower types)
```

Step 2: Split the data into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

Step 3: Create a Decision Tree model

We will use the DecisionTreeClassifier from sklearn

```
model = DecisionTreeClassifier() # Default settings for simplicity
```

Step 4: Train the model

```
model.fit(X_train, y_train)
```

Step 5: Make predictions on the test set

```
y_pred = model.predict(X_test)
```

Step 6: Evaluate the model's accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy of the Decision Tree model: {accuracy * 100:.2f}%")
```

Step 7: Visualize the Decision Tree

Plot the tree using matplotlib

```
plt.figure(figsize=(12, 8))  
plot_tree(model, feature_names=iris.feature_names, class_names=iris.target_names, filled=True)  
plt.title("Decision Tree for Iris Dataset")  
plt.show()
```